

Créer une application **Shopping Cart** complète.

1. Installation

```
npx create-react-app shopping-cart
cd shopping-cart
npm install @reduxjs/toolkit react-redux bootstrap
npm install --save-dev @testing-library/react @testing-library/jest-dom
```

Dans index.js

```
import 'bootstrap/dist/css/bootstrap.min.css';
```

2. Structure du projet :

```
src/
├── app/
│   └── store.js
├── features/cart/
│   ├── cartSlice.js
│   └── cartSlice.test.js
├── components/
│   ├── ProductList.js
│   ├── Cart.js
│   ├── ProductList.test.js
│   └── Cart.test.js
├── App.js
└── index.js
```

3. Store Redux

src/app/store.js

```
import { configureStore } from '@reduxjs/toolkit';
import cartReducer from '../features/cart/cartSlice';

export const store = configureStore({
  reducer: {
    cart: cartReducer
  }
});
```

4. Slice Redux Toolkit (Panier)

src/features/cart/cartSlice.js

```
import { createSlice } from '@reduxjs/toolkit';

const initialState = [];
const cartSlice = createSlice({
  name: 'cart',
  initialState,
```

```

reducers: {
  addToCart(state, action) {
    state.push(action.payload);
  },
  removeFromCart(state, action) {
    return state.filter(item => item.id !== action.payload);
  }
}
});

export const { addToCart, removeFromCart } = cartSlice.actions;
export default cartSlice.reducer;

```

5. Composant ProductList

src/components/ProductList.js

```

import { useDispatch } from 'react-redux';
import { addToCart } from '../features/cart/cartSlice';

const products = [
  { id: 1, name: 'PC', price: 1000 },
  { id: 2, name: 'Téléphone', price: 600 }
];

function ProductList() {
  const dispatch = useDispatch();

  return (
    <div className="container mt-3">
      <h2>Produits</h2>
      {products.map(p => (
        <div key={p.id} className="card mb-2 p-2">
          <strong>{p.name}</strong> - {p.price} €
          <button
            className="btn btn-primary btn-sm float-end"
            onClick={() => dispatch(addToCart(p))}
          >
            Ajouter
          </button>
        </div>
      ))}
    </div>
  );
}

export default ProductList;

```

6. Composant Cart

src/components/Cart.js

```
import { useSelector, useDispatch } from 'react-redux';
import { removeFromCart } from '../features/cart/cartSlice';

function Cart() {
  const cart = useSelector(state => state.cart);
  const dispatch = useDispatch();

  return (
    <div className="container mt-3">
      <h2>Panier</h2>
      <ul className="list-group">
        {cart.map(item => (
          <li key={item.id} className="list-group-item">
            {item.name} - {item.price} €
            <button
              className="btn btn-danger btn-sm float-end"
              onClick={() => dispatch(removeFromCart(item.id))}
            >
              Supprimer
            </button>
          </li>
        ))}
      </ul>
    </div>
  );
}

export default Cart;
```

7. App

src/App.js

```
import ProductList from './components/ProductList';
import Cart from './components/Cart';

function App() {
  return (
    <>
      <ProductList />
      <Cart />
    </>
  );
}

export default App;
```

8. Tests du reducer

src/features/cart/cartSlice.test.js

```
import reducer, { addToCart, removeFromCart } from './cartSlice';

test('ajoute un produit au panier', () => {
  const state = reducer([], addToCart({ id: 1, name: 'PC', price: 1000 }));
  expect(state.length).toBe(1);
});

test('supprime un produit du panier', () => {
  const initialState = [{ id: 1, name: 'PC', price: 1000 }];
  const state = reducer(initialState, removeFromCart(1));
  expect(state.length).toBe(0);
});
```

9. Test ProductList

src/components/ProductList.test.js

```
import { render, screen, fireEvent } from '@testing-library/react';
import { Provider } from 'react-redux';
import { configureStore } from '@reduxjs/toolkit';
import cartReducer from '../features/cart/cartSlice';
import ProductList from './ProductList';

test('ajoute un produit au clic', () => {
  const store = configureStore({ reducer: { cart: cartReducer } });

  render(
    <Provider store={store}>
      <ProductList />
    </Provider>
  );

  fireEvent.click(screen.getAllByText('Ajouter')[0]);
  expect(store.getState().cart.length).toBe(1);
});
```

10. Test Cart

src/components/Cart.test.js

```
import { render, screen } from '@testing-library/react';
import { Provider } from 'react-redux';
import { configureStore } from '@reduxjs/toolkit';
import cartReducer from '../features/cart/cartSlice';
import Cart from './Cart';

test('affiche un produit du panier', () => {
  const store = configureStore({
    reducer: { cart: cartReducer },
    preloadedState: {
      cart: [{ id: 1, name: 'PC', price: 1000 }]
    }
  });

  render(
    <Provider store={store}>
      <Cart />
    </Provider>
  );

  expect(screen.getByText(/PC/)).toBeInTheDocument();
});
```